Unit 8, Lesson 9: Solving Systems by Graphing

Benchmark

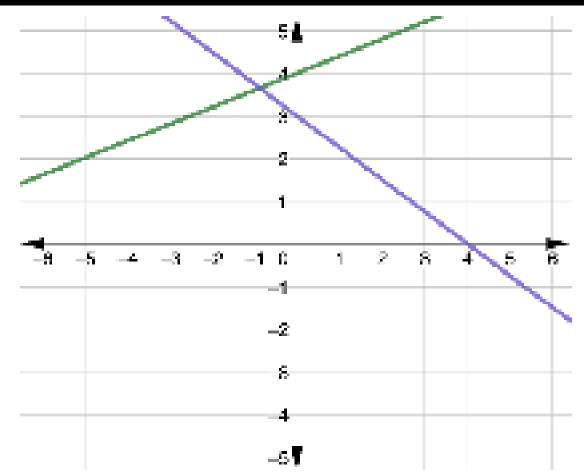
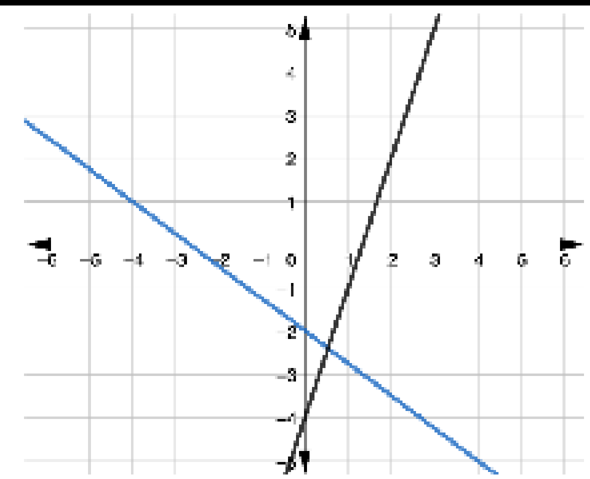
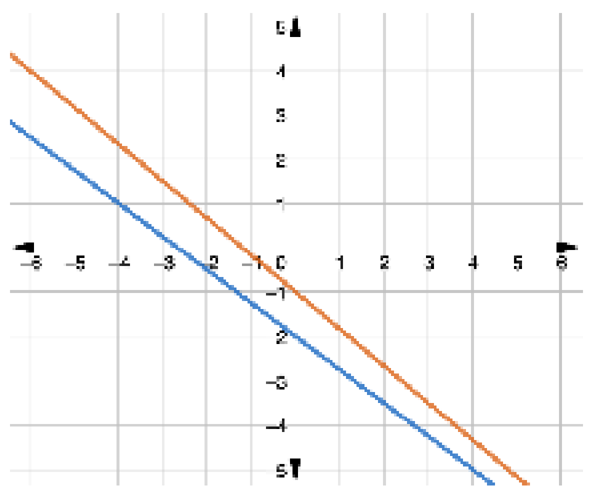
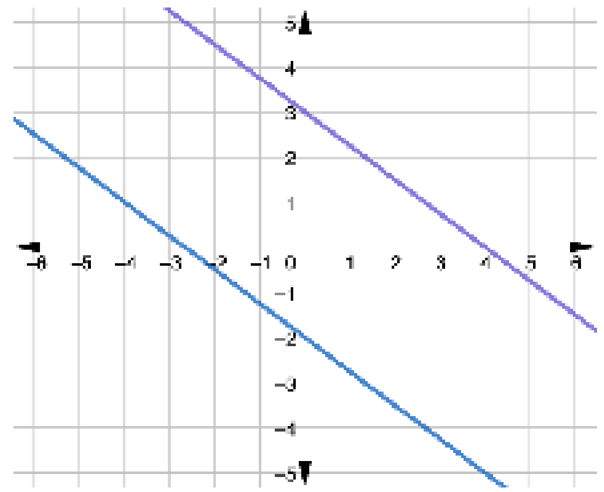
MA.8.AR.4.3 Given a mathematical or real-world context, solve systems of two linear equations by graphing.

Learning Targets

- ☐ I can solve a system of linear equations by graphing.
- ☐ I can verify my answer from graphing by substituting in the x-and y-coordinates of the solution(s).

8.9.1 Warm-Up: Which One Doesn't Belong?

1. The graphs of four systems of equations are shown.

System A	System B
	
System C	**System D**
	

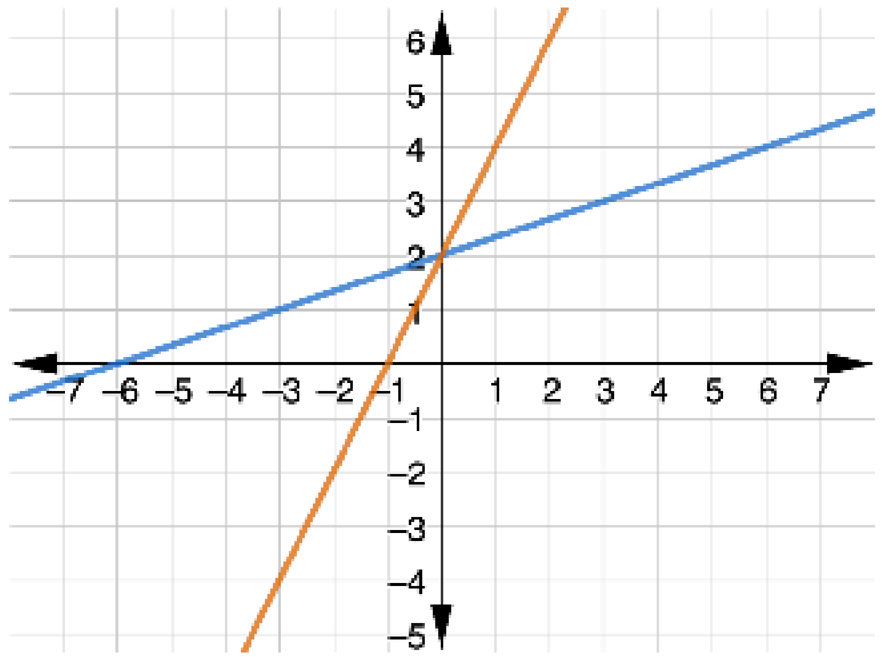
a. Which graph does not belong?

b. Explain your reasoning.

8.9.2 Collaboration: Solving Systems from Graphs

Work with your partner to complete the following.

1. The graph of the system $\begin{cases} y = \frac{1}{3}x + 2 \\ y = 2x + 2 \end{cases}$ is shown.



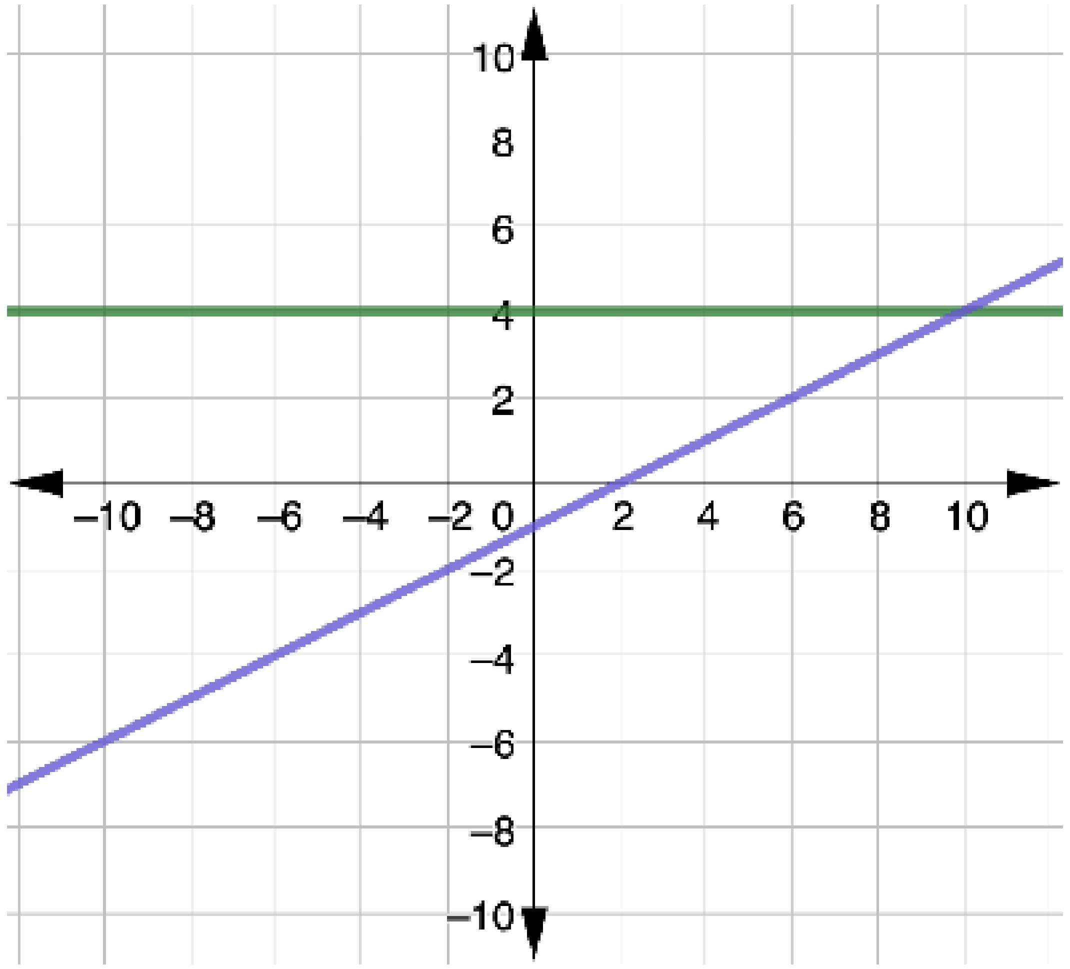
a. Mark the solution to the system on the graph.

b. What are the coordinates of the solution to the system?

c. Verify the solution using substitution.

$y = \frac{1}{3}x + 2$	$y = 2x + 2$

2. The graph of the system $\begin{cases} y = 4 \\ y = \frac{1}{2}x - 1 \end{cases}$ is shown.

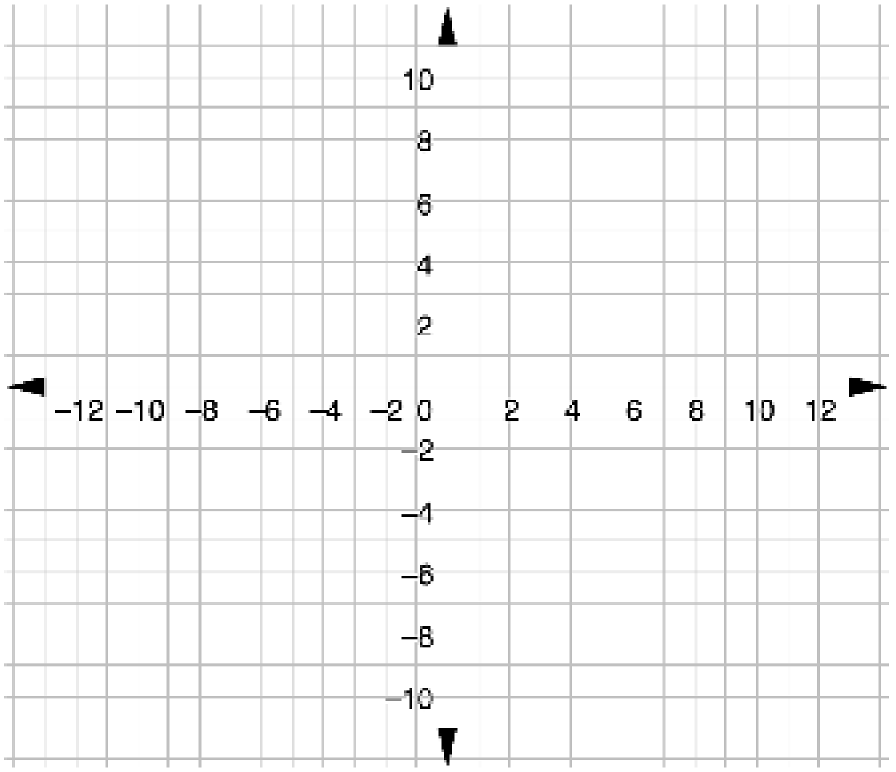


a. Mark the solution to the system on the graph.

b. What are the coordinates of the solution to the system?

c. Verify the solution using substitution.

3. Use the coordinate plane to complete the following.



a. Graph $y = 3x - 3$ on the coordinate plane.

b. Graph $y = \frac{1}{2}x + 2$ on the coordinate plane.

c. Mark and label the solution to the system $\begin{cases} y = 3x - 3 \\ y = \frac{1}{2}x + 2 \end{cases}$ on your graph.

d. Verify your solution using substitution.

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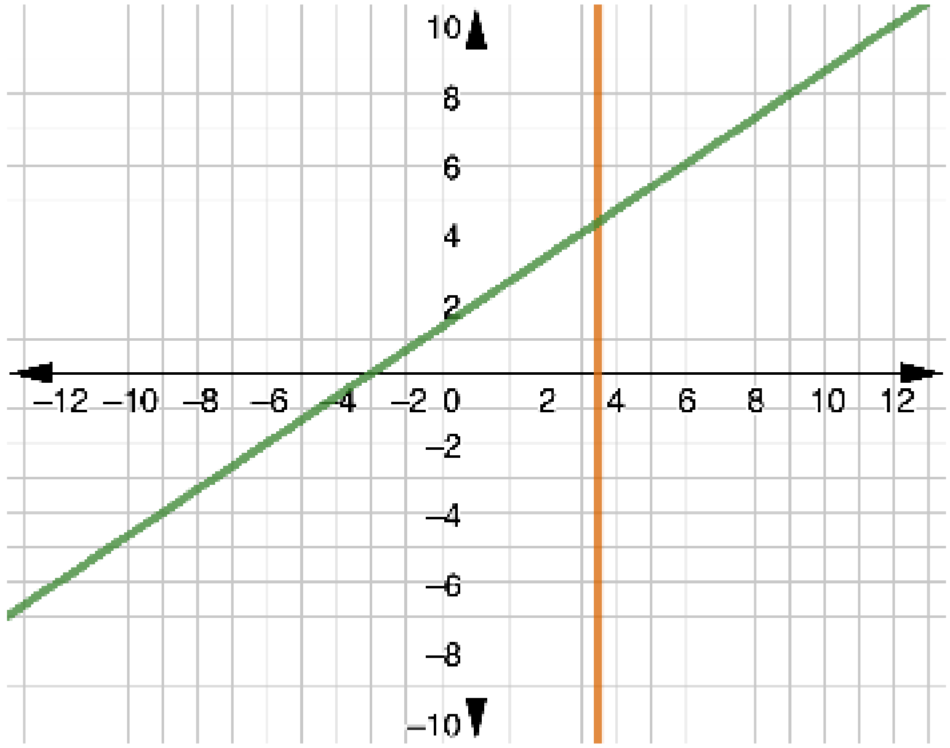
8.9.3 Guided Instruction: Approximating Solutions

1. The graph of the system $\begin{cases} x = 3.5 \\ y = \frac{2}{3}x + 2 \end{cases}$ is shown.



a. Mark the solution to the system of equations.

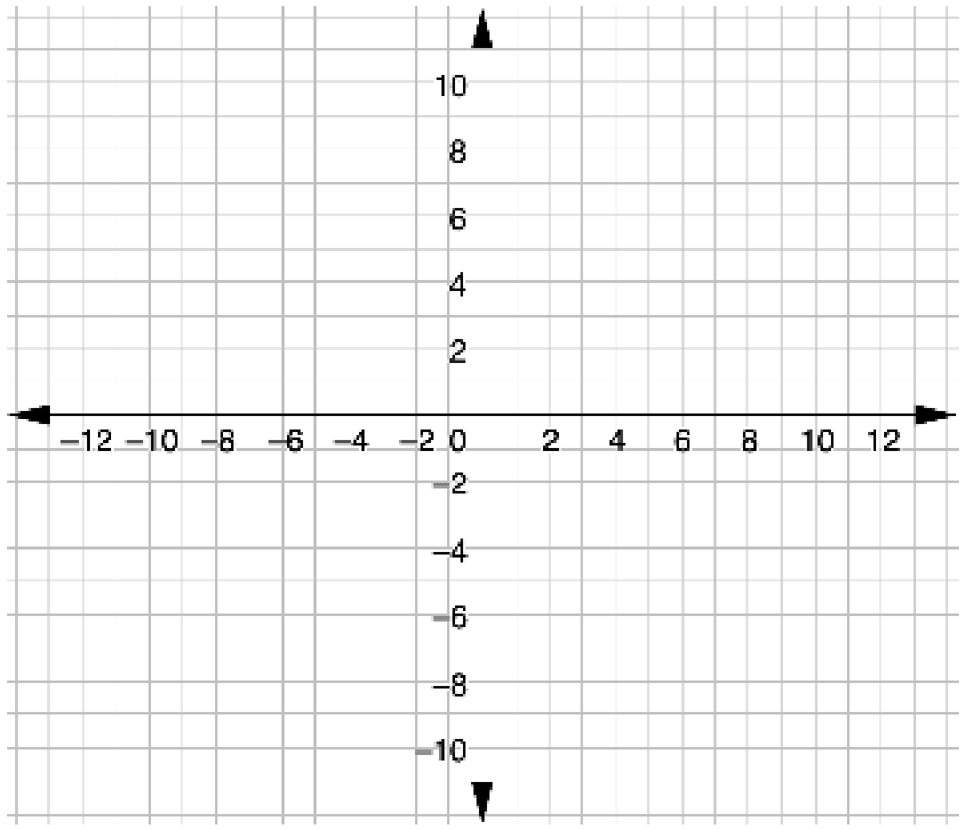
The solution to the system does not fall on the grid lines shown on the coordinate grid. Therefore, the solution can only be approximated from the graph.

b. What are the approximate coordinates of the solution?

c. Kyrie said, "I know the x-value of the solution is 3.5. I could use that to determine the y-value of the solution." Discuss with your partner what Kyrie may have been thinking.

d. Test Kyrie's conjecture to see if his strategy will work to determine the exact solution of the system.

2. Use the coordinate plane to complete the following.



a. Graph the system $\begin{cases} y = -\frac{3}{4}x + 4 \\ y = 3x - 4 \end{cases}$ on the coordinate plane.

b. Mark the solution to the system on your graph.

c. What are the approximate coordinates of the solution?

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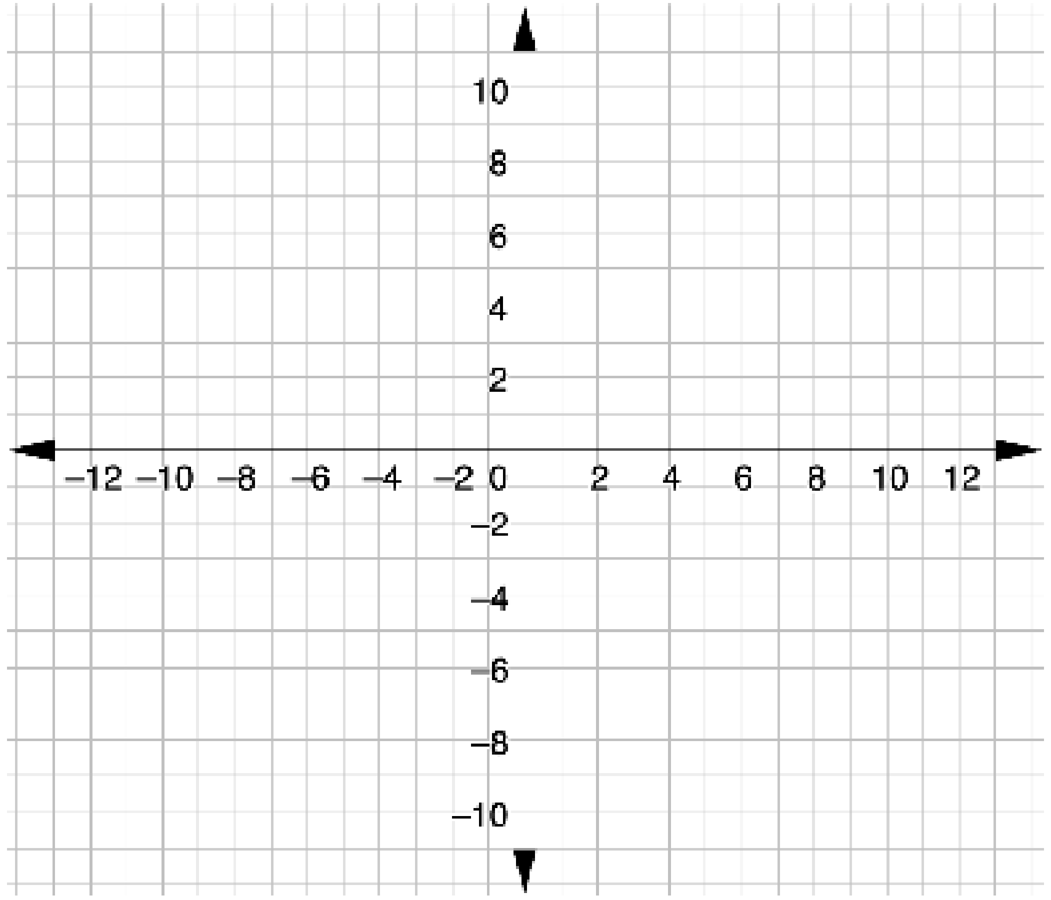
8.9.4 Exploration: Relationships in Parallel Lines

1. **Use the coordinate grid to complete the following.**



a. Graph the system $\begin{cases} y = -\frac{1}{2}x + 4 \\ y = -\frac{1}{2}x - 4 \end{cases}$.

b. Compare your graph with your partner and resolve any differences, if necessary.

c. Discuss with your partner the solution(s) to the system based on your graphs.

d. Work with your partner to determine which statements are true for this system. Then, check all the true statements.

- ☐ The two lines are perpendicular.
- ☐ There is no solution to the system.
- ☐ There are no points that lie on the graph of both lines.
- ☐ There are infinitely many solutions to the system.
- ☐ Any point will make both equations true.
- ☐ The two lines are parallel.

e. Compare the two equations in the system. Discuss with your partner what is the same and what is different about the equations.

f. Complete the statements.

A system of equations that has no solutions occurs when the lines are [o parallel. / o perpendicular.] The equations of [o parallel / o perpendicular] lines have the same [o slopes, / o y-intercepts,] but different [o slopes. / o y-intercepts.]

8.9.5 Bring It Together: Solutions to Systems of Equations

1. The table gives three descriptions of the types of solutions to systems of linear equations. Match each description to the number of solutions the system has.

Description of the System	Number of Solutions
All the ordered pairs that satisfy one equation will also satisfy the other equation.	o one solution o no solutions o infinitely many solutions
Exactly one ordered pair will satisfy both equations. It is the point of intersection of the graphs.	o one solution o no solutions o infinitely many solutions
There are no ordered pairs that will satisfy both equations.	o one solution o no solutions o infinitely many solutions

2. The features of the equations in different types of systems of two linear equations are given. Match each description to the number of solutions the system has.

Description of the System	Number of Solutions
The two lines have **different slopes**.	o one solution o no solutions o infinitely many solutions
The two lines have the **same slope** and the **same *y*-intercept**.	o one solution o no solutions o infinitely many solutions
The two lines have the **same slope** and the **different *y*-intercepts**.	o one solution o no solutions o infinitely many solutions

8.9.6 Wrap-Up: Ticket Out the Door

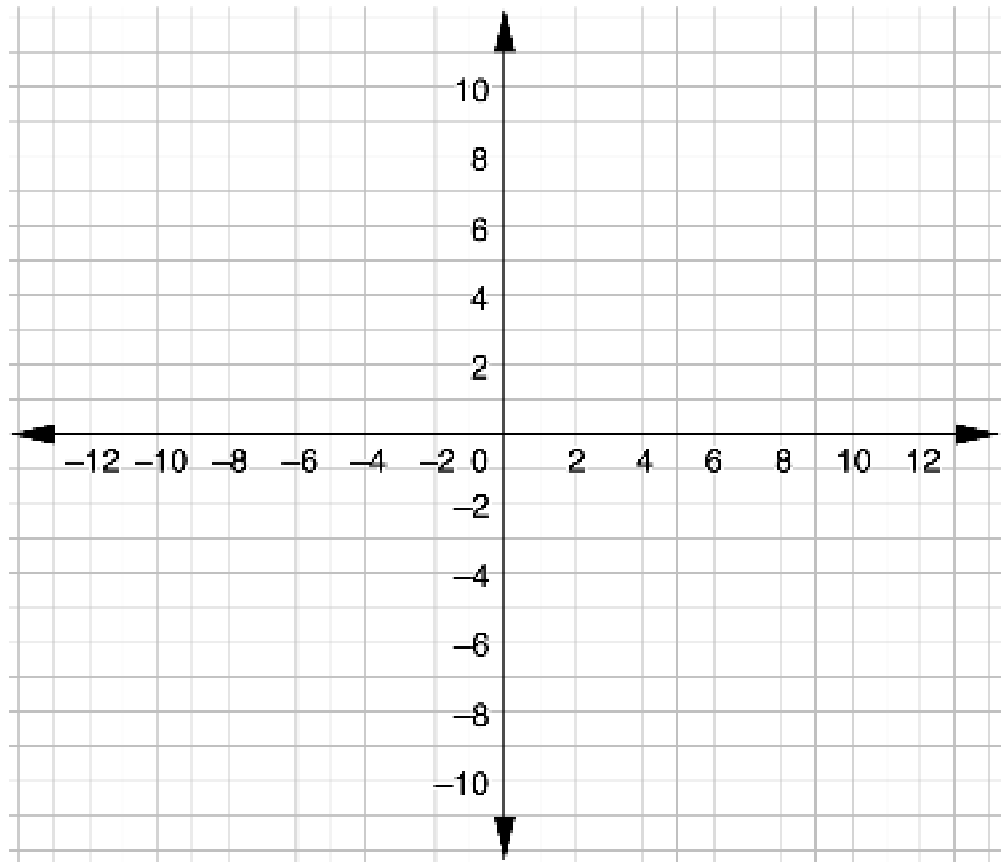
1. Two systems are shown.

$$\begin{cases} y = x + 2 \\ y = 2x - 1 \end{cases} \qquad \begin{cases} y = -x + 3 \\ y = -x - 1 \end{cases}$$

a. Circle the system that will have no solutions.

b. Explain how you know it will have no solutions.

c. Graph the system you did NOT circle in Part A. Then, write the solution under the coordinate plane.



Solution (___, ___)